$$\begin{cases} \dfrac{1}{c^2}\dfrac{\partial^2 \vec{A}}{\partial t^2} - \nabla^2 \vec{A} = 0 \\[2ex] \dfrac{1}{c^2}\dfrac{\partial^2}{\partial t^2}\left(\dfrac{\varphi}{c}\right) - \nabla^2\left(\dfrac{\varphi}{c}\right) = 0 \end{cases}$$

Gauge freedom and fixing.

in vacuum

$$\frac{\partial A^\beta}{\partial x^\beta} = 0 \Rightarrow \frac{1}{c}\frac{\partial}{\partial t}\left(\frac{\varphi}{c}\right) + \vec{\nabla}\cdot\vec{A} = 0$$ Lorentz gauge

let $\begin{cases} \vec{A}' = \vec{A} - \vec{\nabla}\chi : \vec{B}' = \vec{\nabla}\times\vec{A}' = \vec{\nabla}\times\vec{A} = \vec{B} \\ \dfrac{\varphi'}{c} = \dfrac{\varphi}{c} + \dfrac{1}{c}\dfrac{\partial\chi}{\partial t} \quad \vec{E}' = -\dfrac{\partial\vec{A}'}{\partial t} - \vec{\nabla}\varphi' \end{cases}$

$$A'_\alpha = A_\alpha + \frac{\partial\chi}{\partial x^\alpha}$$

$$\vec{E}' = -\frac{\partial\vec{A}}{\partial t} + \frac{\partial}{\partial t}\vec{\nabla}\chi - \left(\vec{\nabla}\varphi + \vec{\nabla}\frac{\partial\chi}{\partial t}\right) =$$

$$= -\frac{\partial\vec{A}}{\partial t} - \vec{\nabla}\varphi + \underbrace{\frac{\partial}{\partial t}\vec{\nabla}\chi - \vec{\nabla}\frac{\partial\chi}{\partial t}}_{=0} = \vec{E}.$$

$$\begin{cases} \dfrac{1}{c^2}\dfrac{\partial^2(\vec{A}' + \vec{\nabla}\chi)}{\partial t^2} - \nabla^2(\vec{A}' + \vec{\nabla}\chi) = 0 \\[2ex] \dfrac{1}{c^2}\dfrac{\partial^2}{\partial t^2}\left(\varphi' - \dfrac{\partial\chi}{\partial t}\right) - \nabla^2\left(\varphi' - \dfrac{\partial\chi}{\partial t}\right) = 0 \end{cases}$$

$$\begin{cases} \dfrac{1}{c^2}\dfrac{\partial^2}{\partial t^2}\vec{A}' - \nabla^2\vec{A}' + \vec{\nabla}\cdot\left[\dfrac{1}{c^2}\dfrac{\partial^2\chi}{\partial t^2} - \nabla^2\chi\right] = 0 \\[2ex] \dfrac{1}{c^2}\dfrac{\partial^2\varphi'}{\partial t^2} - \nabla^2\varphi' - \dfrac{\partial}{\partial t}\left[\dfrac{1}{c^2}\dfrac{\partial^2}{\partial t^2}\chi - \nabla^2\chi\right] = 0 \end{cases}$$

$\Rightarrow$ $\varphi, \vec{A}$ and $\varphi', \vec{A}'$ satisfy the same eqns

if $\frac{1}{c^2}\frac{\partial^2 \chi}{\partial t^2} - \vec{\nabla}^2 \chi = 0$

$\Rightarrow$ we can choose χ such that:

$$\varphi' = \varphi + \frac{\partial \chi}{\partial t} = 0 \quad ! \Rightarrow \vec{\nabla} \cdot \vec{A}' + \frac{1}{c^2}\frac{\partial}{\partial t}\varphi' = 0$$

$$\Rightarrow \boxed{\vec{\nabla} \cdot \vec{A}' = 0} \text{ Coulomb gauge}$$

$\Rightarrow$ need only one potential, $\vec{A}$, to describe EM field in free space $\rightarrow$ convenient for field quantization.

Quantization of $\vec{A}$

$$\vec{\nabla}\cdot\vec{A}=0 \quad , \quad \nabla^2\vec{A}-\frac{1}{c^2}\frac{\partial^2\vec{A}}{\partial t^2}=0$$

let us place field in the box of volume $V=L^3$

PBC $\quad \vec{A}(\vec{r},t)=\vec{A}(\vec{r}+L\hat{e}_j,t)$

Fourier decomposition

$$\vec{A}(\vec{r},t)=\sum_{\vec{k}\,\sigma} u_{k\sigma}(t)\,\vec{e}_{k\sigma}\,A_0\,e^{i\vec{k}\cdot\vec{r}}$$

↳ normalization factor

$$\vec{\nabla}\cdot\vec{A}=i\sum_{k\sigma}\vec{k}\cdot\vec{e}_{k\sigma}\,u_{k\sigma}(t)\,A_0\,e^{i\vec{k}\cdot\vec{r}}=0$$

$\Rightarrow$ polarization vectors $\hat{e}_{k\sigma}\cdot\vec{k}=0$ are orthogonal to $\vec{k}$ $(\sigma=1,2)$ $\vec{e}_1\cdot\vec{e}_2=0$.

(transverse wave)

PBC gives $e^{ik_xL}=e^{ik_yL}=e^{ik_zL}=1$

$$\vec{k}=\frac{2\pi}{L}\vec{n} \quad (n_x,n_y,n_z)=\vec{n}$$

$$\left(\vec{A}(\vec{r},t)\right)^*=\vec{A}(\vec{r},t) \quad \text{(real)}$$

$$\parallel$$

$$\sum_{k\sigma} u^*_{k\sigma}(t)\,\vec{e}_{k\sigma}\,A_0^*\,e^{-i\vec{k}\cdot\vec{r}} = (\vec{k}\to-\vec{k})$$

→ make real

$$=\sum_{k\sigma} u^*_{-k\sigma}(t)\,\vec{e}_{-k\sigma}\,A_0^*\,e^{i\vec{k}\cdot\vec{r}}$$

$$\Rightarrow u_{\vec{k}\sigma}=u^*_{-\vec{k}\sigma} \quad (\text{for } \vec{e}_{\vec{k}\,1,2}\sim\hat{x} \text{ or } \hat{y})$$

Note that $\int d^3\vec{r}\, e^{i(\vec{k}-\vec{k}')\cdot\vec{r}} =$

$$= \underbrace{\int_0^L dx\, e^{ix(n_x-n'_x)\frac{2\pi}{L}}}_{\begin{cases} L & \text{for } n_x = n'_x \\ 0 & \text{for } n_x \neq n'_x \end{cases}} \int dy(\;)\int dz(\;) = L^3 \delta_{\vec{n}\vec{n}'}$$

$$\vec{\nabla}^2 \vec{A}(\vec{r},t) = \sum_{k\sigma} u_{k\sigma}\, \hat{e}_{k\sigma} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right) e^{i\vec{k}\cdot\vec{r}}$$

$$= -\sum_{k\sigma} u_{k\sigma}\, \hat{e}_{k\sigma} (\vec{k}^2)\, e^{i\vec{k}\cdot\vec{r}}$$

$$\frac{\partial^2}{\partial t^2} \vec{A} = \sum_{k\sigma} \ddot{u}_{k\sigma}(t)\, \hat{e}_{k\sigma}\, e^{i\vec{k}\cdot\vec{r}}$$

$$\Rightarrow \sum_{\vec{k}\sigma} \left(\frac{1}{c^2}\ddot{u}_{\vec{k}\sigma} + k^2 u_{\vec{k}\sigma}\right) \hat{e}_{k\sigma}\, e^{i\vec{k}\cdot\vec{r}} = 0$$

Multiply by $e^{-i\vec{q}\cdot\vec{r}}$ and integrate over $\vec{r}$:

$$\begin{cases} \ddot{u}_{\vec{q}\sigma}(t) + (c\vec{q})^2 u_{\vec{q}\sigma}(t) = 0, & \omega_{\vec{q}} \equiv c|\vec{q}| = cq. \\ \dfrac{\partial^2 u_{q\sigma}}{\partial t^2} = -\omega_q^2 u_{q\sigma} & \text{Harmonic oscillators, } \underline{\text{for every } \vec{q} \text{ and } \sigma} \end{cases}$$

General solution

$$u_{\vec{k}\sigma}(t) = b_{\vec{k}\sigma}\, e^{-i\omega_k t} + c_{\vec{k}\sigma}\, e^{i\omega_k t}$$

complex coefficients (independent of t)

Reality condition: $\vec{A}(\vec{r},t) = \vec{A}(\vec{r},t)^*$

$$\sum_{\vec{k}\sigma} \underbrace{\left(b_{\vec{k}\sigma}\, e^{-i\omega_k t} + c_{\vec{k}\sigma}\, e^{i\omega_k t}\right)}_{=u_{\vec{k}\sigma}} \hat{e}_{\vec{k}\sigma}\, e^{i\vec{k}\cdot\vec{r}} =$$

$$= \sum_{\vec{k}\sigma} \underbrace{\left(b^*_{-\vec{k}\sigma}\, e^{i\omega_k t} + c^*_{-\vec{k}\sigma}\, e^{-i\omega_k t}\right)}_{=u^*_{-\vec{k}\sigma}} \hat{e}^*_{-\vec{k}\sigma}\, e^{i\vec{k}\cdot\vec{r}}$$

$$\Rightarrow \begin{cases} \hat{e}_{k\sigma}\, b_{\vec{k}\sigma} = \hat{e}^*_{-\vec{k}\sigma}\, c^*_{-\vec{k}\sigma} \\ \hat{e}_{k\sigma}\, c_{\vec{k}\sigma} = \hat{e}^*_{-\vec{k}\sigma}\, b^*_{-\vec{k}\sigma} \end{cases}$$

Let us choose polarization vectors so that

$$\hat{e}_{k\sigma} = \hat{e}^*_{-k\sigma} \quad \Rightarrow \quad \hat{e}^*_{\vec{k}\sigma} = \hat{e}_{-\vec{k}\sigma}$$

$$\Rightarrow \begin{cases} b_{\vec{k}\sigma} = c^*_{-\vec{k}\sigma} \\ c_{\vec{k}\sigma} = b^*_{-\vec{k}\sigma} \end{cases}$$

$$\Rightarrow u_{\vec{k}\sigma} = b_{\vec{k}\sigma}\, e^{-i\omega_k t} + b^*_{-\vec{k}\sigma}\, e^{i\omega_k t}$$

$$\vec{A}(\vec{r},t) = \sum_{\vec{k}\sigma} \left(b_{\vec{k}\sigma}\, e^{-i\omega_k t} + b^*_{-\vec{k}\sigma}\, e^{i\omega_k t}\right) \hat{e}_{\vec{k}\sigma}\, e^{i\vec{k}\cdot\vec{r}}$$

$$= \sum_{\vec{k}\sigma} b_{\vec{k}\sigma}\, e^{-i\omega_k t + i\vec{k}\cdot\vec{r}}\, \hat{e}_{\vec{k}\sigma} + \sum_{(\vec{k}\to-\vec{k})} b^*_{\vec{k}\sigma}\, e^{i\omega_k t - i\vec{k}\cdot\vec{r}}\, \underbrace{\hat{e}_{-\vec{k}\sigma}}_{\hat{e}^*_{\vec{k}\sigma}}$$

$$\Rightarrow \quad \vec{A}(\vec{r},t) = \sum_{\vec{k}\sigma} \Big\{ b_{\vec{k}\sigma}\, e^{-i\omega_k t + i\vec{k}\cdot\vec{r}}\, \hat{e}_{\vec{k}\sigma}$$

$$+ \left(b_{\vec{k}\sigma}\, e^{-i\omega_k t + i\vec{k}\cdot\vec{r}}\, \hat{e}_{\vec{k}\sigma} \right)^* \Big\} (A_0).$$

Field energy density:

$$E = \frac{\varepsilon_0}{2} \vec{E}^2 + \frac{1}{2\mu_0} \vec{B}^2 = \frac{\varepsilon_0}{2} \left(\vec{E}^2 + c^2 \vec{B}^2 \right)$$

$$c^2 = \frac{1}{\varepsilon_0 \mu_0} \qquad \vec{E} = -\frac{\partial \vec{A}}{\partial t}, \quad \vec{B} = \vec{\nabla} \times \vec{A} = \hat{r}^a \varepsilon^{abc} \frac{\partial}{\partial x^b} A^c$$

$$= \hat{r}^a \varepsilon^{abc}\, i k^b A^c$$

$$= i\, \vec{k} \times \vec{A}.$$

$$\vec{E} = \sum_{\vec{k}\sigma} \Big\{ i\omega_k\, b_{\vec{k}\sigma}\, e^{-i\omega_k t + i\vec{k}\cdot\vec{r}}\, \hat{e}_{\vec{k}\sigma}$$

$$- i\omega_k\, b^*_{\vec{k}\sigma}\, e^{i\omega_k t - i\vec{k}\cdot\vec{r}}\, \hat{e}^*_{\vec{k}\sigma} \Big\}$$

$$\vec{B} = \sum_{\vec{k}\sigma} \left(i\, \vec{k} \times \hat{e}_{\vec{k}\sigma}\, b_{\vec{k}\sigma}\, e^{-i\omega_k t + i\vec{k}\cdot\vec{r}} + i\, \vec{k} \times \hat{e}^*_{\vec{k}\sigma}\, b^*_{\vec{k}\sigma}\, e^{i\omega_k t - i\vec{k}\cdot\vec{r}} \right)$$

In volume V:

$$E = -\frac{\varepsilon_0}{2} \int d^3\vec{r} \sum_{\vec{k}\sigma} \sum_{\vec{k}'\sigma'} \Big\{ \omega_k \left(b_{\vec{k}\sigma}\, e^{-i\omega_k t + i\vec{k}\vec{r}}\, \hat{e}_{\vec{k}\sigma} - b^*_{\vec{k}\sigma}\, e^{i\omega_k t - i\vec{k}\vec{r}}\, \hat{e}^*_{\vec{k}\sigma} \right)$$

$$\cdot\, \omega_{k'} \left(b_{k'\sigma'}\, e^{-i\omega'_k t + ik'r}\, \hat{e}_{k'\sigma'} - b^*_{k'\sigma'}\, e^{i\omega'_k t - ik'r}\, \hat{e}^*_{k'\sigma'} \right)$$

$$+ c^2 \left(\vec{k} \times \hat{e}_{k\sigma}\, b_{k\sigma}\, e^{-i\omega_k t + ik\cdot r} + \vec{k} \times \hat{e}^*_{k\sigma}\, b^*_{k\sigma}\, e^{i\omega_k t - i\vec{k}\cdot\vec{r}} \right)$$

$$\left(\vec{k}' \times \hat{e}_{k'\sigma'}\, b_{k'\sigma'}\, e^{-i\omega'_k t + ik'r} + \vec{k}' \times \hat{e}^*_{k'\sigma'}\, b^*_{k'\sigma'}\, e^{i\omega'_k t - ik'r} \right) \Big\}$$

$$\int d^3\vec{r}\, e^{i\vec{r}\cdot(\vec{k}+\vec{k}')} = L^3\, \delta_{\vec{k}',-\vec{k}}$$

$$\int d^3\vec{r}\, e^{i\vec{r}\cdot(\vec{k}-\vec{k}')} = L^3\, \delta_{\vec{k}',\vec{k}}$$

$b_{k\sigma} b_{k'\sigma'}$ terms: $\sum_{\substack{\vec{k}\sigma \\ \vec{k}'\sigma'}} \delta_{\vec{k}',-\vec{k}} \; b_{k\sigma} b_{-k\sigma'} \Big[e^{-i2\omega_k t} \hat{e}_{k\sigma} \cdot \hat{e}_{-k\sigma'} \omega_k^2$

$$+ c^2 (\vec{k} \times \hat{e}_{k\sigma}) \cdot (-\vec{k} \times \hat{e}_{-k\sigma'}) \; e^{-i2\omega_k t} \Big] = 0$$

$$(\vec{k} \times \hat{e}_{k\sigma}) \cdot (\vec{k} \times \hat{e}_{-k\sigma'}) = \vec{k} \cdot (\hat{e}_{-k\sigma'} \times \vec{k} \times \hat{e}_{k\sigma}) =$$

$$\underset{A}{\vec{k}} \cdot \Big(\underset{B \times C}{\vec{k} \; (\hat{e}_{-k\sigma'} \cdot \hat{e}_{k\sigma}) - \hat{e}_{k\sigma} \underbrace{(\vec{k} \cdot \hat{e}_{-k\sigma'})}_{\emptyset \text{ (transverse wave)}}} \Big) =$$

$$= k^2 (\hat{e}_{k\sigma} \cdot \hat{e}_{-k\sigma'})$$

The whole bb thing $= 0$ because $\omega_k^2 - c^2 k^2 = 0$.

$\Rightarrow$ only $\delta_{\vec{k}',\vec{k}}$ term contributes

$$E = \frac{\varepsilon_0}{2} \sum_{\vec{k}\sigma\sigma'} \omega_k^2 \left(b_{k\sigma} b^*_{k\sigma'} \hat{e}_{k\sigma} \cdot \hat{e}^*_{k\sigma'} + b^*_{k\sigma} b_{k\sigma'} \hat{e}^*_{k\sigma} \cdot \hat{e}_{k\sigma'} \right)$$

$$+ c^2 \Big[(\vec{k} \times \hat{e}_{k\sigma}) \cdot (\vec{k} \times \hat{e}^*_{k\sigma'}) \, b_{k\sigma} b^*_{k\sigma'}$$

$$+ (\vec{k} \times \hat{e}^*_{k\sigma}) \cdot (\vec{k} \times \hat{e}_{k\sigma'}) \, b^*_{k\sigma} b_{k\sigma'} \Big]$$

$$= \frac{\varepsilon_0 L^3}{2} \sum_{\vec{k}\sigma\sigma'} (\omega_k^2 + c^2 k^2) \Big(\hat{e}_{k\sigma} \cdot \hat{e}^*_{k\sigma'} \; b^*_{k\sigma'} b_{k\sigma}$$

$$+ \underbrace{\hat{e}^*_{k\sigma} \cdot \hat{e}_{k\sigma'}}_{\delta_{\sigma\sigma'}} \; b^*_{k\sigma} b_{k\sigma'} \Big)$$

$$= 2\varepsilon_0 V \sum_{\vec{k}\sigma} \omega_k^2 \; b^*_{k\sigma} b_{k\sigma}$$

let $b_{k\sigma} = \frac{1}{2}\left(q_{k\sigma} + \frac{i}{\omega_k} p_{k\sigma}\right)$ with real $q_{k\sigma}, p_{k\sigma}$

$$E = 2\varepsilon_0 V \sum_{k\sigma} \omega_k^2 \, \frac{1}{2}\left(q_{k\sigma} - \frac{i}{\omega_k} p_{k\sigma}\right)\left(\frac{1}{2}\right)\left(q_{k\sigma} + \frac{i}{\omega_k} p_{k\sigma}\right)$$

$$= \frac{1}{2}\varepsilon_0 V \sum_{k\sigma}\left(\omega_k^2 q_{k\sigma}^2 + p_{k\sigma}^2\right)$$

(At this point can choose overall normalization $A_0^2 = \frac{1}{\varepsilon_0 V}$)

Then $E \to \mathcal{H} = \sum_{k\sigma} \mathcal{H}_{k\sigma} = \sum_{k\sigma} \frac{1}{2}\left(p_{k\sigma}^2 + \omega_k^2 q_{k\sigma}^2\right)$ SHO with unit mass

Eq. of motion:
$$\begin{cases} \dot{q}_{k\sigma} = \dfrac{\partial \mathcal{H}_{k\sigma}}{\partial p_{k\sigma}} = p_{k\sigma} \\ \dot{p}_{k\sigma} = -\dfrac{\partial \mathcal{H}_{k\sigma}}{\partial q_{k\sigma}} = -\omega_k^2 q_{k\sigma} \end{cases}$$

$$\ddot{p}_{k\sigma} = -\omega_k^2 \dot{q}_{k\sigma} = -\omega_k^2 p_{k\sigma} \quad \text{etc.}$$

Thus $\mathcal{H} = \sum_{\vec{k}\sigma} \{$ energy of harmonic oscillators for each $(\vec{k},\sigma)\}$

Quantize $\hat{\mathcal{H}} = \sum_{k\sigma} \frac{1}{2}\left(\hat{p}_{k\sigma}^2 + \omega_k^2 \hat{q}_{k\sigma}^2\right)$

operators $[\hat{q}_{k\sigma}, \hat{p}_{k\sigma}] = i\hbar$

$[\hat{q}_{k\sigma}, \hat{p}_{k'\sigma'}] = i\hbar\,\delta_{kk'}\delta_{\sigma\sigma'}$

let $\hat{a}^+_{k\sigma} = \frac{1}{\sqrt{2\hbar\omega_k}}\left(-i\,\hat{p}_{k\sigma} + \omega_k\,\hat{q}_{k\sigma}\right)$

$$\hat{a}_{k\sigma} = \frac{1}{\sqrt{2\hbar\omega_k}}\left(i\,\hat{p}_{k\sigma} + \omega_{k\sigma}\,\hat{q}_{k\sigma}\right)$$

Such that $[\hat{a},\hat{a}^+] = \frac{1}{2\hbar\omega_k}\Big(\underbrace{[i\hat{p},\omega\hat{q}]}_{i\omega(-i\hbar)} - \underbrace{[\omega\hat{q}, i\hat{p}]}_{i\omega(i\hbar)}\Big) =$

$$= \frac{1}{2\hbar\omega}\,2\omega\hbar = 1.$$

$$\hat{p}_{k\sigma} = \sqrt{2\hbar\omega_k}\,\frac{\hat{a}_{k\sigma} - \hat{a}^+_{k\sigma}}{2i}, \quad \hat{q}_{k\sigma} = \sqrt{2\hbar\omega_k}\,\frac{\hat{a}_{k\sigma} + \hat{a}^+_{k\sigma}}{2\omega_k}$$

$$\frac{1}{2}\left(\hat{p}^2_{k\sigma} + \omega_k^2\,\hat{q}^2_{k\sigma}\right) = \frac{1}{2}\Bigg(\frac{\hbar\omega_k}{2}\,\frac{a-a^+}{i}\,\frac{a-a^+}{i} +$$

$$+\omega_k^2\,\frac{\hbar}{2\omega_k}\,(a+a^+)(a+a^+)\Bigg) = \frac{1}{4}\hbar\omega\Big(-(aa + a^+a^+ - aa^+ - a^+a)$$

$$+\ aa + a^+a^+ + aa^+ + a^+a\Big) = \frac{1}{2}\hbar\omega\,(aa^+ + a^+a) =$$

$$= \frac{1}{2}\hbar\omega(2a^+a + 1) = \hbar\omega\left(a^+a + \frac{1}{2}\right)$$

$$\Rightarrow \quad \mathcal{H} = \sum_{k\sigma} \hbar\omega_k\left(\hat{a}^+_{k\sigma}\hat{a}_{k\sigma} + \frac{1}{2}\right) \ !$$

$$\begin{cases} \hat{q}_{k\sigma} = \sqrt{\frac{\hbar}{2\omega_k}}\,(\hat{a}_{k\sigma} + \hat{a}^+_{k\sigma}) \\ \hat{p}_{k\sigma} = i\sqrt{\frac{\hbar\omega_k}{2}}\,(\hat{a}^+_{k\sigma} - \hat{a}_{k\sigma}) \end{cases}$$

Work backwards now:

$$\vec{A}(\bar{r},t) = A_0 \sum_{\bar{k}\sigma} b_{k\sigma}\, e^{-i\omega_k t + i\bar{k}\cdot\bar{r}}\, \hat{e}_{k\sigma} + b^*_{k\sigma}\, e^{i\omega_k t - i\bar{k}\cdot\bar{r}}\, \hat{e}^*_{k\sigma}$$

$$= \frac{A_0}{2} \sum_{k\sigma} \left(q_{k\sigma} + \frac{i}{\omega_k} p_{k\sigma}\right) e^{-i\omega_k t + i\bar{k}\cdot\bar{r}}\, \hat{e}_{k\sigma} +$$

$$+ \left(q_{k\sigma} - \frac{i}{\omega_k} p_{k\sigma}\right) e^{i\omega_k t - i\bar{k}\cdot\bar{r}}\, \hat{e}^*_{k\sigma} =$$

$$= \frac{A_0}{2} \sum_{k\sigma} \sqrt{\frac{\hbar}{2\omega_k}} (a^+ + a)\left(e^{-i\omega t + i\bar{k}\bar{r}}\, \hat{e}_{k\sigma} + e^{i\omega t - i\bar{k}\bar{r}}\, \hat{e}^*_{k\sigma}\right)$$

$$+ \frac{i\cdot i}{\omega_k} \sqrt{\frac{\hbar\omega_k}{2}} (a^+ - a)\left(e^{-i\omega t + i\bar{k}\bar{r}}\, \hat{e}_{k\sigma} - e^{i\omega t - i\bar{k}\bar{r}}\, \hat{e}^*_{k\sigma}\right);$$

But $b_{k\sigma} = \frac{1}{2}\left(q_{k\sigma} + \frac{i}{\omega_k} p_{k\sigma}\right) = \frac{1}{2}\left[\sqrt{\frac{\hbar}{2\omega_k}}(a + a^+) + \frac{i}{\omega_k}\, i\sqrt{\frac{\hbar\omega_k}{2}}(a^+ - a)\right]$

$$= \frac{1}{2}\sqrt{\frac{\hbar}{2\omega_k}}\left(a + a^+ - (a^+ - a)\right) = \sqrt{\frac{\hbar}{2\omega_k}}\, \hat{a}_{k\sigma}$$

$$b^*_{k\sigma} = \sqrt{\frac{\hbar}{2\omega_k}}\, \hat{a}^+_{k\sigma}$$

$$\Rightarrow \vec{A}(\vec{r},t) = \sum_{\vec{k}\sigma} \sqrt{\frac{\hbar}{2\varepsilon_0 V \omega_k}} \left\{ \hat{a}_{k\sigma}\, e^{-i\omega_k t + i\bar{k}\cdot\bar{r}}\, \hat{e}_{k\sigma} + \hat{a}^+_{k\sigma}\, e^{i\omega_k t - i\bar{k}\cdot\bar{r}}\, \hat{e}^*_{k\sigma} \right\}$$

<u>(quantized EM field)</u>

$$\vec{A}(\vec{r},t) = \sum_{\vec{k}\sigma} \hat{a}_{k\sigma}\, \vec{A}_{k\sigma}(\bar{r},t) + \hat{a}^+_{k\sigma}\, \vec{A}^*_{k\sigma}(\bar{r},t)$$

where $\vec{A}_{k\sigma}(\vec{r},t) = \hat{e}_{k\sigma}\sqrt{\frac{\hbar}{2\varepsilon_0 V \omega_k}}\, e^{-i\omega_k t + i\vec{k}\cdot\vec{r}}$

$$\vec{E}(\vec{r},t) = -\frac{\partial \vec{A}}{\partial t} = \sum_{\vec{k}\sigma} i\omega_k \left(\hat{a}_{k\sigma} \vec{A}_{k\sigma} - \hat{a}^+_{k\sigma} \vec{A}^*_{k\sigma} \right)$$

$$\vec{B}(\vec{r},t) = \vec{\nabla} \times \vec{A} = \sum_{\vec{k}\sigma} i \vec{k} \times \left(\hat{a}_{k\sigma} \vec{A}_{k\sigma} - \hat{a}^+_{k\sigma} \vec{A}^*_{k\sigma} \right)$$

$$\mathcal{H} = \sum_{\vec{k}\sigma} \hbar\omega_k \left(\underbrace{\hat{a}^+_{k\sigma} \hat{a}_{k\sigma}}_{\hat{N}_{k\sigma}} + \frac{1}{2} \right)$$

(ideal gas of photons!)

$\hat{N}_{k\sigma}$ = number operator of the EM mode $(\vec{k}, \sigma)$

$\Rightarrow \quad \Psi = \Psi\{\hat{N}_{k\sigma}\} = \sum_{\vec{k}\sigma} |N_{k\sigma}\rangle$ describes the eigenstate

$$a^+ |n\rangle = \sqrt{n+1}\, |n+1\rangle$$

$$a |n\rangle = \sqrt{n}\, |n-1\rangle$$

$$a^+ a |n\rangle = n |n\rangle$$

$$\left(a^+ a |n\rangle = a^+ \sqrt{n}\, |n-1\rangle = \sqrt{n}\, \sqrt{n-1+1}\, |n-1+1\rangle = n|n\rangle \right).$$

$\Rightarrow$ thermodynamics of photon gas:

Gibbs distribution $W_s = \frac{1}{Z} e^{-\varepsilon_s / k_B T}$

$$\varepsilon_s = \hbar\omega_s \left(N_s + \frac{1}{2} \right)$$

$$\overline{N} = \frac{1}{Z}\sum_s N_s\, e^{-\frac{\hbar\omega}{k_B T}\left(N_s + \frac{1}{2}\right)} = \frac{1}{Z}\sum_{s=0}^{\infty} n\, e^{-xn}\, e^{-x/2}$$

$$Z = \sum_n e^{-x\left(n+\frac{1}{2}\right)} = \frac{e^{-x/2}}{1-e^{-x}}$$

$$\overline{N} = \frac{\sum_{n=0}^{\infty} n\, e^{-xn}}{\sum_{n=0}^{\infty} e^{-xn}} = \frac{+e^{-x}}{\left(1-e^{-x}\right)^2}\cdot\left(1-e^{-x}\right) = \frac{-e^{-x}}{e^{-x}-1} =$$

$$= \frac{1}{e^{x}-1}$$

$$\sum n\, e^{-xn} = -\frac{\partial}{\partial x}\sum e^{-xn} = -\frac{\partial}{\partial x}\,\frac{1}{1-e^{-x}} =$$

$$= +\frac{e^{-x}}{\left(1-e^{-x}\right)^2}$$

Quantum theory of radiation + matter

Ham. of charged particle in EM field

$$\hat{\mathcal{H}}_{part} = \frac{1}{2m}\left(\hat{p} - q\vec{A}\right)^2 + q\varphi - \underbrace{\vec{\mu}\cdot\vec{B}}_{\text{spin mag. moment}} \longrightarrow$$

$$\rightarrow \frac{\hat{p}^2}{2m} - \frac{q}{2m}\left(\hat{p}\cdot\vec{A} + \vec{A}\cdot\hat{p}\right) + \frac{q^2}{2m}\vec{A}^2 + q\varphi - \vec{\mu}\cdot\vec{B}$$

$$\hat{p} = -i\hbar\vec{\nabla} \Rightarrow \hat{p}\cdot\vec{A} \approx -i\hbar\frac{\partial}{\partial\vec{r}}\left(e^{i\vec{k}\cdot\vec{r}}\hat{e}_{k\sigma}\right) =$$

$$= -i\hbar\left(\vec{k}\cdot\hat{e}_{k\sigma}\right)e^{i\vec{k}\cdot\vec{r}} = 0$$

$$\Rightarrow \hat{p}\cdot\vec{A} = \vec{A}\cdot\hat{p}$$

Main interaction term $\left(-\frac{q}{m}\vec{A}\cdot\hat{p}\right) = \hat{V}$
(single photon process)

$\vec{A}^2 \Rightarrow$ leads to 2-photon processes

Separate $e^{\pm i\omega_k t}$ factors from $\vec{A}_{k\sigma}(\vec{r},t) = e^{-i\omega_k t}\vec{A}_{k\sigma}(\vec{r})$

Use perturbation theory:

probability of transition between states $|i\rangle$ and $|f\rangle$

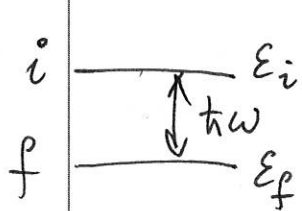
$$dw = \frac{2\pi}{\hbar}\left|\langle f|\hat{V}|i\rangle\right|^2 \underbrace{\delta(\varepsilon_i - \varepsilon_f - \hbar\omega_k)}_{\frac{1}{\hbar}\delta\left(\frac{\varepsilon_i - \varepsilon_f - \omega}{\hbar}\right)} \frac{V k^2 dk\, d\Omega_k}{(2\pi)^3}$$

number of different states (with fixed polarization σ) in interval $d^3\vec{k}$.

$\vec{k}$-space: distance between adjacent $\vec{k}$-states

$$\Delta k_j = \frac{2\pi}{L}$$

$\Rightarrow$ # states in volume $d^3\vec{k}$ is $dN = \frac{d^3\vec{k}}{\left(\frac{2\pi}{L}\right)^3} = \frac{L^3 d^3\vec{k}}{(2\pi)^2}$.



<u>Emission:</u> $|i\rangle = \psi_i \, \phi(\{N\})$

$$|f\rangle = \psi_f \, \phi(\{N+1\})$$

These are atomic states $\psi_{i,f}$.

$$\Rightarrow \langle f | \hat{V} | i \rangle = -\frac{q}{m} \langle \phi_{N+1} | \int d^3\vec{r} \, \langle \psi_f | \hat{p} | \psi_i \rangle \cdot \vec{A} \, | \phi_N \rangle$$

$$\int d^3r \, \langle \psi_f | \hat{p} | \psi_i \rangle = -i\hbar \int d^3\vec{r} \, \psi_f^* (\vec{\nabla} \psi_i) =$$

$$= -i\hbar \int d^3\vec{r} \left\{ \vec{\nabla} (\psi_f^* \psi_i) - (\vec{\nabla} \psi_f^*) \psi_i \right\} =$$

$$= i\hbar \int d^3\vec{r} \, \vec{\nabla} \psi_f^* \, \psi_i$$

$$\Rightarrow \cancel{} \quad + \frac{q}{m} \langle \psi_f | \hat{p} | \psi_i \rangle = \underbrace{\frac{-i\hbar q}{2m} \left\{ \langle \psi_f | \vec{\nabla} \psi_i \rangle - \langle \vec{\nabla} \psi_f | \psi_i \rangle \right\}}_{\text{current operator } (q=e)}$$

<u>Dipole radiation</u>

$$\vec{A}_{k\sigma} = \hat{e}_{k\sigma} \sqrt{\frac{\hbar}{2\varepsilon_0 V \omega_k}} \, e^{i\vec{k}\cdot\vec{r}} \approx \hat{e}_{k\sigma} \sqrt{\frac{\hbar}{2\varepsilon_0 V \omega_k}}$$

because $\vec{k} \cdot \vec{r} \approx 2\pi \frac{r}{\lambda} \ll 1$ (where r is dipole's distribution)

$$\int d^3\vec{r} \, \langle \psi_f | \underbrace{\frac{-i\hbar \vec{\nabla}}{m}}_{\hat{v} \,=\, \text{velocity operator}} | \psi_i \rangle$$

$$\hat{v} = \dot{\vec{r}} = \frac{i}{\hbar}\left[\hat{\mathcal{H}}, \vec{r}\right] = \frac{i}{\hbar}\left(\hat{\mathcal{H}}\,\hat{r} - \hat{r}\,\hat{\mathcal{H}}\right)$$

$$\langle \psi_f | \hat{v} | \psi_i \rangle = \frac{i}{\hbar} \langle \psi_f | \hat{\mathcal{H}}\hat{r} - \hat{r}\hat{\mathcal{H}} | \psi_i \rangle =$$

$$= \frac{i}{\hbar}\left(\varepsilon_f - \varepsilon_i\right) \langle \psi_f | \hat{r} | \psi_i \rangle$$

$$\langle f | \hat{v} | i \rangle = \frac{-iq}{\hbar}\left(\varepsilon_f - \varepsilon_i\right) \int d^3\vec{r}\, \langle \psi_f | \hat{r} | \psi_i \rangle \cdot$$

$$\sum_{\vec{k}\sigma} \cdot \hat{e}^*_{k\sigma} \sqrt{\frac{\hbar}{2\varepsilon_0 V \omega_k}} \underbrace{\langle \phi_{N+1} | \hat{a}^+_{k\sigma} | \phi_N \rangle}_{\sqrt{N_k+1}}$$

$$= -\frac{i}{\hbar} \sum_{\sigma} \hat{d}_{fi} \cdot \hat{e}^*_{k\sigma} \hbar \sqrt{\frac{\hbar\omega_k}{2\varepsilon_0 V}} \sqrt{N_k+1}$$

where $\hbar\omega_k = \varepsilon_f - \varepsilon_i$ selects $\vec{k}$.

$$\cancel{dW = \frac{2\pi}{\hbar}\left|\sum_\sigma \hat{e}^*_{k\sigma} \cdot \hat{d}_{fi}\right|^2}$$

Here $\hat{d}_{fi} = q \int d^3\vec{r}\, \langle \psi_f | \hat{r} | \psi_i \rangle$

dipole matrix element between states $|f\rangle$ and $|i\rangle$

$$dW = \frac{2\pi}{\hbar}\left|\frac{-i}{\hbar}\sum_{\vec{k}\sigma}(\varepsilon_f-\varepsilon_i)\,\hat{d}_{fi}\cdot\hat{e}^*_{k\sigma}\sqrt{\frac{\hbar}{2\varepsilon_0 V\omega_k}}\sqrt{N_k+1}\right|^2$$

$$\cdot\ \frac{V\omega^2 d\omega\, d\Omega_k}{(2\pi c)^3}\ \underbrace{\delta(\varepsilon_i-\varepsilon_f-\hbar\omega)}_{\frac{1}{\hbar}\delta\left(\frac{\varepsilon_i-\varepsilon_f}{\hbar}-\omega\right)}$$

$$\Rightarrow W=\int dW = \frac{2\pi}{\hbar^2}\ \frac{\hbar\omega_k}{2\varepsilon_0 V}(N_k+1)\ \frac{V\omega^2 d\Omega_k}{(2\pi c)^3}\left|\sum_\sigma \hat{e}^*_{k\sigma}\cdot\hat{d}_{fi}\right|^2$$

$$W_{fi} = \frac{\pi}{\varepsilon_0}(N_\omega+1)\frac{\omega^3}{8\pi^3c^3\hbar}|\vec{d}\times\hat{n}|^2\, d\Omega_k$$

$$\int d\Omega_k\ \sin^2\theta = 2\pi\int_0^\pi d\theta\,\sin\theta\,\sin^2\theta =$$

$$= 2\pi\int_{-1}^{1} d(\cos\theta)\,(1-\cos^2\theta) = 4\pi\int_0^1 dx(1-x^2) = 4\pi\left(1-\frac{1}{3}\right)=\frac{8\pi}{3}$$

$$W_{fi} = (N_\omega+1)\ \frac{\omega^3|d_{fi}|^2}{3\pi c^3\varepsilon_0\hbar}$$

$$\frac{1}{c^2}=\varepsilon_0\mu_0$$

$$\text{Emitted energy} = W_{fi}\cdot\hbar\omega = (N_\omega+1)\frac{\mu_0\omega^4|\hat{d}_{fi}|^2}{3\pi c}$$

$$\left|\sum_\sigma \hat{e}^*_{k\sigma}\cdot\hat{d}\right| = \left(\hat{e}^*_1+\hat{e}^*_2\right)\cdot\hat{d}\,\left(\hat{e}_1+\hat{e}_2\right)\hat{d}^*$$

$$= \left(\hat{e}^*_{1a}+\hat{e}^*_{2a}\right)\hat{d}_a\left(\hat{e}_{1b}+\hat{e}_{2b}\right)\hat{d}^*_b$$

$$\hat{e}^*_i\cdot\hat{e}_j=\delta_{ij}\quad,\quad \hat{n}\cdot\hat{e}_i=0$$

Let us expand unit vectors $\varepsilon_a = t_1^{(1)}\hat{e}_1 + t_2^{(2)}\hat{e}_2 + t^{(3)}\hat{n}$

$$\varepsilon_a\cdot\varepsilon_b=\delta_{ab}=\left(t_a^{(1)}\hat{e}_1+t_a^{(2)}\hat{e}_2+t_a^{(3)}\hat{n}\right)\cdot$$

$$\cdot\left(t_b^{(1)}\hat{e}_1+t_b^{(2)}\hat{e}_2+t_b^{(3)}\hat{n}\right)$$

$$= t_a^{(1)}t_b^{(1)}+t_a^{(2)}t_b^{(2)}+\underbrace{t_a^{(3)}t_b^{(3)}}_{n_a n_b}$$

$$t_\gamma^{(3)}=n_\gamma$$

$$\Rightarrow\ \hat{e}_a^{(\sigma)}\hat{e}_b^{(\sigma)}=\delta_{ab}-n_a n_b$$

$$\Rightarrow\ \sum_{\sigma\sigma'}\hat{e}_{\sigma a}\,\hat{e}_{\sigma' b}=\delta_{ab}-n_a n_b$$

$$\Rightarrow\ \sum_{\sigma\sigma'}\sum_{ab}\hat{e}_{\sigma a}\,\hat{e}_{\sigma' b}\,\hat{d}_a\,\hat{d}_b=\sum_{ab}\left(\delta_{ab}-n_a n_b\right)d_a d_b$$

$$=\vec{d}\cdot\vec{d}-\left(\vec{d}\cdot\hat{n}\right)^2=\left|\vec{d}\times\hat{n}\right|^2.$$